In Class Assignment (Graded)

Python In-Class Assignment: Lists, Sets, Tuples, and Dictionaries

Instructions:

- Write your code in a Python file
- Each question should be solved one by one.
- Add comments (#) in your code to explain what you are doing.
- Assignment is graded
- submit the single python file with all the code in one single file.
- Submit the assignment before the class on Monday July 27th.

Part 1: Lists

1. Create a list called `fruits` with the following values: `"apple"`, `"banana"`, `"cherry"`, `"mango"`.
2. Print the second item in the list.
3. Add `"orange"` to the list.
4. Change the first item of the list to `"grape"`.
5. Print the length of the list.

Part 2: Sets

1. Create a set called `colors` with the values: `"red"`, `"blue"`, `"green"`, `"yellow"`.
2. Add `"purple"` to the set.

3. Try adding `"red"` again and print the set. What do you observe?

4. Remove `"blue"` from the set.

Part 3: Tuples

1. Create a tuple called `days` with values: `"Monday", "Tuesday", "Wednesday", "Thursday", "Friday"`.

2. Print the third item from the tuple.

3. Try changing `"Tuesday"` to `"Tues"` and see what error you get.

4. Use slicing to print the first three items of the tuple.

Part 4: Dictionaries

1. Create a dictionary called `student` with keys: `"name"`, `"age"`, `"grade"` and values: `"Alex"`, `20`, `"A"`.

2. Print the student's name.

3. Add a new key `"city"` with value `"Toronto"`.

4. Change the `"grade"` to `"B+"`.

5. Print all the keys and values using a loop.

Part 5: If/Else Statements (Conditionals)

1. Create a variable called `temperature` and set it to 25.

2. Write an `if/elif/else` block that checks the temperature:

    - If it is greater than 30, print "It's a hot day."

    - If it is between 15 and 30 (inclusive), print "It's a beautiful day."

    - If it is less than 15, print "It's a cold day."

3. Create a boolean variable called `is_raining` and set it to `True`. Write an `if/else` statement that prints "Bring an umbrella" if true, and "Leave the umbrella at home" if false.

Part 6: For Loops

1. Create a list called `numbers` containing the integers `1, 2, 3, 4, 5`.

2. Write a `for` loop that iterates through the `numbers` list and prints each number multiplied by 2.

Part 7: While Loops

1. Create a variable called `counter` and set it to 0.

2. Write a `while` loop that prints the value of `counter` as long as it is less than 5. (Don't forget to add 1 to the `counter` inside the loop so it doesn't run forever!)

3. Create a new `while` loop that starts a variable at 10 and subtracts 1 each time. Use an `if` statement and the `break` keyword to stop the loop completely when the variable reaches 5. Print a "Loop ended!" message at the end.

0 % 0 of 1 topics complete

In class assignment 2 (Graded)

Assignment